

Vinitkumar Jani

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SOFTWARE ENGINEER PROFILE

High-performing Computing Science undergraduate (Ranked 1st in Department) at Coventry University. Technically proficient in **Full-Stack Development** using **Python (Flask), Java, and SQL**. Expert in **Object-Oriented Programming (OOP)** and **Data Structures**, with a proven ability to optimize backend systems for **40% faster performance**. Strong focus on building secure, scalable applications and automating workflows.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), Java, SQL (MySQL/PostgreSQL), JavaScript, HTML5, CSS3, **Base64 Frame Transmission**.
 - **Frameworks & Libraries:** Flask, **FastAPI (Asynchronous API)**, SQLAlchemy (ORM), NumPy, Pandas, **ONNX Runtime**, **Dlib (Face Recognition)**.
 - **Core CS Concepts:** Data Structures & Algorithms (DSA), OOP, **Decoupled Architecture**, **NMS-Free End-to-End Prediction**, RESTful APIs, System Design.
 - **AI & DevOps:** **YOLOv26 (Real-time Object Detection)**, **Transfer Learning**, Git/GitHub, Linux CLI, ETL Pipelines, **Model Optimization (PT to ONNX)**.
 - **Security:** **OWASP Top 10 Mitigation**, Input Sanitization, Secure Session Management, **Biometric Privacy by Design**.
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SOFTWARE ENGINEERING PROJECTS

VigilAI: AI-Driven Security Full-Stack Artefact

- **Backend & API:** Architected a decoupled system using **FastAPI** to manage high-speed inference requests via **Base64** frame transmission .
- **System Design:** Optimized for portability and speed using a **JSON-based flat-file architecture** and **ONNX Runtime**, eliminating the need for heavy database management systems.
- **Frontend:** Developed a high-performance dashboard using **Tailwind CSS and Vanilla JS** featuring real-time detection rendering and automated weapon alerts.
- **Optimization:** Leveraged **NMS-Free end-to-end prediction** to reduce CPU inference latency by eliminating the Non-Maximum Suppression bottleneck.

London Crime Analytics Engine | Python, MySQL, SQLAlchemy

- **Backend Engineering:** Architected a **MySQL Star Schema** to manage 13M+ records, ensuring ACID compliance and relational integrity.
- **ETL Pipeline:** Developed a Python-based automated data pipeline using **SQLAlchemy** for efficient extraction and transformation.
- **Performance:** Optimized query execution by implementing **SQL Indexes** and **CTEs**, reducing system latency by **40%**.
- **Visual Integration:** Integrated a Seaborn-based plotting engine to generate dynamic system reports on urban safety trends.

MediTrack: Hospital Management System | Python, MySQL, Tkinter

- **Full-Stack Development:** Built a desktop application for healthcare administration, featuring patient record management and real-time appointment scheduling.
- **Algorithmic Optimization:** Integrated **QuickSort** and **MergeSort** algorithms to index and retrieve data, achieving a **40% speed increase** over standard search methods.
- **GUI Design:** Developed a user-friendly interface using Tkinter, ensuring seamless interaction between the frontend and the MySQL backend.

Journey Advisor: Full-Stack Travel Planner | Python (Flask), MySQL

- **Web Development:** Engineered a full-stack web app following the **MVC pattern**, calculating carbon footprints and journey costs for UK travel.
- **Security & UX:** Implemented secure session management, user authentication, and a dynamic QR-code ticket generation system.

PROFESSIONAL EXPERIENCE

Victoria Solutions | London *Software & Data Analysis Intern* | Jul 2025 – Sep 2025

- **System Automation:** Developed Python scripts to automate complex data transformation tasks, improving reporting speed for executive teams.
- **Technical Excellence:** Awarded **Certificate of Excellence** for high-quality technical standards and software documentation.

G Asian Canteen | London *IT Systems Manager* | Jun 2024 – Aug 2024

- **Solution Architect:** Digitalized a manual inventory process by building a custom digital tracking system, increasing **data accuracy by 20%**.
- **Efficiency Gain:** Optimized IT workflows and inventory logic, leading to a **25% improvement** in overall operational throughput.

Pho Vietnamese Street Food | London *Server / Kitchen Operations* | Jan 2024 – Jan 2026

- **Operational Logic:** Optimized high-volume kitchen workflows and managed real-time order processing via POS systems during peak demand.

EDUCATION

BSc Computing Science | Coventry University, London

- **Status:** Ranked 1st in the Department of Computing.
 - **Modules:** Algorithms & Programming (Python/Java), Software Engineering, Data Science & Databases, IT Infrastructure.
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CERTIFICATIONS

- **Learning SQL Programming** – LinkedIn & NASBA (2026).
- **Data & AI Internship Certificate** – Victoria Solutions (2025).
- **IEUK Internship Experience** – Bright Network (2025).